



Chapter 1.

1.

DEC	OCT	HEX
50 ₁₀	= 62 ₈	= 32 ₁₆
51 ₁₀	= 63 ₈	= 33 ₁₆
52 ₁₀	= 64 ₈	= 34 ₁₆
53 ₁₀	= 65 ₈	= 35 ₁₆
54 ₁₀	= 66 ₈	= 36 ₁₆
55 ₁₀	= 67 ₈	= 37 ₁₆
56 ₁₀	= 70 ₈	= 38 ₁₆
57 ₁₀	= 71 ₈	= 39 ₁₆

DEC	OCT	HEX
58 ₁₀	= 72 ₈	= 3A ₁₆
59 ₁₀	= 73 ₈	= 3B ₁₆
60 ₁₀	= 74 ₈	= 3C ₁₆
61 ₁₀	= 75 ₈	= 3D ₁₆
62 ₁₀	= 76 ₈	= 3E ₁₆
63 ₁₀	= 77 ₈	= 3F ₁₆
64 ₁₀	= 80 ₈	= 40 ₁₆

2.

<key: 2¹⁰ = K, 2²⁰ = M, 2³⁰ = G, 2⁴⁰ = T, PPT 9>

(a) 16K bytes = $16 \times 2^{10} = 16384$

(b) 32M bytes = $32 \times 2^{20} = 33554432$

(c) 2G bytes = $2 \times 2^{30} = 2147483648$

3.

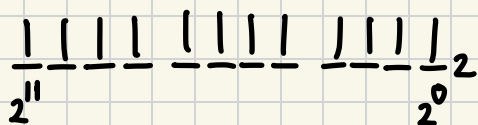
(a) $1203_4 = 1 \times 4^3 + 2 \times 4^2 + 0 \times 4^1 + 3 \times 4^0 = 99_{10}$

(b) $5243_6 = 5 \times 6^3 + 2 \times 6^2 + 4 \times 6^1 + 3 \times 6^0 = 1179_{10}$

(c) $9922_{14} = 9 \times 14^3 + 9 \times 14^2 + 2 \times 14^1 + 2 \times 14^0 = 26490_{10}$

(d) $248_9 = 2 \times 9^2 + 4 \times 9^1 + 8 \times 9^0 = 206_{10}$

4.



$$\text{DEC} = 2^{12} - 1 = 4096 - 1 = 4095$$

$$\text{OCT} = 4095_{10} = 7777_8$$

$$\text{HEX} = 4095_{10} = \text{FFF}_{16}$$

5.

$$(a) 67 \div 5 = 11 \text{ R } 13 \quad (b) 15 \times 3 = 51 \quad (c) 123 + 120 = 303$$

$$67_b = 6b + 7$$

$$15_b = b + 5$$

$$123 = b^2 + 2b + 3$$

$$5_b = 5$$

$$3_b = 3$$

$$120 = b^2 + 2b$$

$$13_b = b + 3$$

$$51_b = 5b + 1$$

$$303 = 3b^2 + 3$$

$$6b + 7 = 5b + 15$$

$$3b + 15 = 5b + 1$$

$$3b^2 + 4b + 3 = 3b^2 + 3$$

$$b = 8$$

$$2b = 14, b = 7$$

$$b^2 - 4b = 0$$

$$b(b - 4) = 0, b = 4 \text{ or } 0$$

Ans: 4

6.

$$x_b^2 - 13x_b + 22b = 0$$

$$x = 7 \text{ or } 2$$

$$13_b = b + 3, 22_b = 2b + 2$$

$$\Rightarrow x^2 - (b + 3)x + (2b + 2) = 0$$

$$(x - 7)(x - 2) = 0$$

$$\Rightarrow x^2 - 9x + 14$$

$$x^2 - (b + 3)x + (2b + 2) = x^2 - 9x + 14$$

$$b + 3 = 9, b = 6$$

$$2b + 2 = 14, 2b = 12, b = 6$$

Ans: $b = 6$

7.

C	A	5	E
(12)	(110)	(5)	(14)
1100	1010	0101	1110
001	100	101	001
1	4	5	1
			3
			6

CAE₁₆

$$= (1 \ 100 \ 101 \ 001 \ 011 \ 110)_2$$

$$= (145136)_8$$

8.

(a)

$$\begin{array}{r}
 2 \overline{) 253} \\
 \underline{2} \\
 2 \overline{) 26} \\
 \underline{2} \\
 2 \overline{) 63} \\
 \underline{2} \\
 2 \overline{) 3} \\
 \underline{2} \\
 2 \overline{) 15} \\
 \underline{2} \\
 2 \overline{) 7} \\
 \underline{2} \\
 2 \overline{) 3} \\
 \underline{2} \\
 1
 \end{array}$$

$$253_{10} = 1111101_2$$

(b)

$$\begin{array}{r}
 16 \overline{) 253} \\
 \underline{16} \\
 15
 \end{array}$$

$$253_{10} = FD_{16} = 1111101_2$$

b more faster than a.

9.

$$\begin{aligned}
 (a) \ 10101.101_2 &= 2^4 + 2^2 + 2^0 + 2^{-1} + 2^{-3} = 16 + 4 + 1 + 0.5 + 0.125 \\
 &= 21.625_{10}
 \end{aligned}$$

$$(b) \ 64.8_{16} = 6 \times 16^1 + 4 \times 16^0 + 8 \times 16^{-1} = 96 + 4 + 0.5 = 100.5_{10}$$

$$\begin{aligned}
 (c) \ 261.44_8 &= 2 \times 8^2 + 6 \times 8^1 + 1 \times 8^0 + 4 \times 8^{-1} + 4 \times 8^{-2} \\
 &= 128 + 48 + 1 + 0.5 + 0.625 = 178.125_{10}
 \end{aligned}$$

$$\begin{aligned}
 (d) \ 51DE.C_{16} &= 5 \times 16^3 + 1 \times 16^2 + 13 \times 16^1 + 14 \times 16^0 + 12 \times 16^{-1} \\
 &= 20480 + 256 + 208 + 14 + 0.75 \\
 &= 20958.75_{10}
 \end{aligned}$$

$$\begin{aligned}
 (e) \ 110011.001_2 &= 32 + 16 + 2 + 1 + 0.125 = 51.125_{10} \\
 142.125_{10}
 \end{aligned}$$

10.

$$(a) 1.00011 = 1 \times 2^0 + 1 \times 2^{-4} + 1 \times 2^{-5} = 1 + 0.0625 + 0.03125 \\ = (1.09375)_{10}$$

$$(1.09375)_{10} = 1 + 0.58 = (1.58)_{16}$$

$$0.09375 \times 16 = 1.5$$

$$0.5 \times 16 = 8.0$$

$$(b) 1000.11 = 1 \times 2^3 + 1 \times 2^{-1} + 1 \times 2^{-2} = 8.75$$

$$(8.75)_{10} = 8 + 0.C = (8.C)_{16}$$

$$0.75 \times 16 = 12 (C)$$

因為 (b) 的 bin 相較於 (a) 向右移了三格，
也就是 $2^3 = 8$ ，所以 (a) · (b) 相差 8 倍。

11.

$$101010 = 2^5 + 2^3 + 2^1 = 32 + 8 + 2 = 42$$

$$100 = 2^2 = 4$$

$$42 \div 4 = 10.5$$

$$(10.5)_{10} = (1010)_2 + (0.1)_2 = 1010.1 \neq$$

$$0.5 \times 2 = 1.0$$

12.

(a) $1101 + 110 = 10011$ (b) $D0 + 1F = (EF)_{16}$
 $1101 \times 110 = 1001110$ $D0 \times 1F = (1930)_{16}$

1101	1101
+ 110	x 110
-----	-----
10011	0000
	1101
	1101

	1001110

D0	D0
+ 1F	x 1F
-----	-----
EF	C30
	D0

	1930

13.

(a) $45.125 = 101101.001_2$ (b) $3.33 = (11)_2 + (0.0101)_2$

$45 = 32 + 8 + 4 + 1$
 $= 2^5 + 2^3 + 2^2 + 2^0$
 $= (101101)_2$

$0.125 \times 2 = 0.25 \quad 0$
 $0.25 \times 2 = 0.5 \quad 0$
 $0.5 \times 2 = 1.0 \quad 1$

$0.33 \times 2 = 0.66 \quad 0$
 $0.66 \times 2 = 1.32 \quad 1$
 $0.32 \times 2 = 0.64 \quad 0$
 $0.64 \times 2 = 1.28 \quad 1$

$(11.0101)_2 = 2^1 + 2^0 + 2^{-2} + 2^{-4}$
 $= 3 + 0.25 + 0.0625$
 $= 3.3125$
 $\frac{3.33 - 3.3125}{3.33} \times 100\% \approx 0.525\%$

(c) $11.0101_2 = 3.3125_{10} = 3 + 0.5 = (3.5)_{16}$
 $0.3125 \times 16 = 5$
 $(3.5)_{16} = 3 + 5 \times 16^{-1} = 3.3125_{10}$, Not Same.

14.

	1's Complement	2's Complement
11100010	00011101	00011110
00011000	11100111	11101000
10111101	01000010	01000011
10100101	01011010	01011011
11000011	00111100	00111101
01011000	10100111	10101000

15.

	9's	10's
6523 4035	3476 5964	3476 5965
5678 3223	4321 6776	4321 6777
8700 0367	1299 9632	1299 9633
9999 9000	0000 9999	0000 1000

16.

$$(a) (2360)_8 \xrightarrow{7's} 5417 \xrightarrow{8's} 5420$$

$$(b) (2360)_8 = 01001110000$$

$$(c) 1's \Rightarrow 101100001111 \xrightarrow{2's} \underline{101} \underline{1000} \underline{10000}$$

$$(d) \frac{101}{5} \quad \frac{100}{4} \quad \frac{010}{2} \quad \frac{000}{0}, \text{ Same as (a).}$$

17.

$$(a) 7523 - 4.567$$

$$9999 - 4.567 + 1 = 9995.433$$

$$\text{Ans: } 7518.433$$

$$\begin{array}{r} 7523 \\ + 9995.433 \\ \hline 17518.433 \\ - 10000.433 \\ \hline 7518.433 \end{array}$$

$$(b) 230 - 1204$$

$$9999 - 1204 + 1 = 8796$$

$$- (9999 - 9026 + 1) = -974^*$$

$$\text{Ans: } -974$$

$$\begin{array}{r} 0230 \\ + 8796 \\ \hline 9026 \end{array}$$

$$(c) 224 - 712$$

$$999 - 712 + 1 = 288$$

$$- (999 - 512 + 1) = -488^*$$

$$\text{Ans: } -488$$

$$\begin{array}{r} 224 \\ + 288 \\ \hline 512 \end{array}$$

$$(d) 2390 - 945$$

$$9999 - 945 + 1 = 9055$$

$$\text{Ans: } 1445$$

$$\begin{array}{r} 2390 \\ + 9055 \\ \hline 11445 \\ - 10000 \\ \hline 1445 \end{array}$$

18.

(a) $11001 - 10010$

$10010 \xrightarrow{1's} 01101 \xrightarrow{2's} 01110$

Ans: 00111

$$\begin{array}{r} 11001 \\ + 01110 \\ \hline 100111 \\ - 100000 \\ \hline 00111 \end{array}$$

(b) $1100 - 111100$

$111100 \xrightarrow{1's} 000011 \xrightarrow{2's} 000100$

$001000 \xrightarrow{1's} 110111 \xrightarrow{2's} 111000$

Ans: -111000

$$\begin{array}{r} 11 \\ 001100 \\ + 111100 \\ \hline 1001000 \\ - 1000000 \\ \hline 001000 \end{array}$$

(c) $10101 - 11011$

$11011 \xrightarrow{1's} 00100 \xrightarrow{2's} 00101$

$11010 \xrightarrow{1's} 00101 \xrightarrow{2's} 00110$

Ans: -00110

$$\begin{array}{r} 10101 \\ + 00101 \\ \hline 11010 \end{array}$$

(d) $1100011 - 10001$

$0010001 \xrightarrow{1's} 1101110 \xrightarrow{2's} 1101111$

Ans: 1010010

$$\begin{array}{r} 1100011 \\ + 1101111 \\ \hline 11010010 \\ - 10000000 \\ \hline 1010010 \end{array}$$

19.

$$(a) (+9286) + (+801)$$

$$= 09286 + 00801 = 10087$$

$$\begin{array}{r} 09286 \\ + 00801 \\ \hline 10087 \end{array}$$

$$(b) (+9286) + (-801)$$

$$801 \xrightarrow{9's} 99198 \xrightarrow{10's} 99199$$

$$\text{Ans} : 8485$$

$$\begin{array}{r} 09286 \\ + 99199 \\ \hline 108485 \\ - 100000 \\ \hline 8485 \end{array}$$

$$(c) (-9286) + (+801)$$

$$9286 \xrightarrow{9's} 990713 \xrightarrow{10's} 990714 \quad 990714$$

$$\text{Ans} : 991515 \#$$

$$\begin{array}{r} + 000801 \\ \hline 991515 \end{array}$$

$$(d) (-9286) + (-801)$$

$$9286 \xrightarrow{9's} 990713 \xrightarrow{10's} 990714$$

$$801 \xrightarrow{9's} 999198 \xrightarrow{10's} 999199$$

$$\text{Ans} : 989913$$

$$\begin{array}{r} 990714 \\ + 999199 \\ \hline 1989913 \\ - 1000000 \\ \hline 989913 \end{array}$$

24.

5	2	1	1	DEC
0	0	0	0	0
0	0	0	1	1
0	0	1	1	2
0	1	0	1	3
0	1	1	1	4
1	0	0	0	5
1	0	0	1	6
1	0	1	1	7
1	1	0	1	8
1	1	1	1	9

29.

<u>100 0011</u>	<u>010 1110</u>	<u>100 0010</u>	<u>100 0001</u>	<u>100 0010</u>
C	.	B	A	B
<u>100 0010</u>	<u>100 0111</u>	<u>100 0101</u>		
B	G	E		

31.

< pptx P. 42 >

94 printing characters.

34 special characters.

32.

第6位元。

33. 1000 011 0101 1001

(a) 4 DEC in BCD

(BCD=8421)

8 7 5 9

(b) 4 DEC in Excess-3

5 4 2 6

Excess3:

$0_{10} : 0011$

$9_{10} : 1100$

开始, 接不斷 $+1$, 直到形成閉環。
 -1

(c) 4 HEX in Gray Codes F 5 6 E

(d) 2 ASCII BEL ; Y